JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON SUNDAY 05th APRIL 2026) **TIME : 3:00 PM TO 6:00 PM**

PHYSICS

TEST PAPER WITH SOLUTION

SECTION-A

26. Match List-I with List-II.

List-I	List-II
(A) Meter (L)	I. $\sqrt{\frac{hc}{G}}$
(B) Second (S)	II. $\sqrt{\frac{Gh}{c^5}}$
(C) Kilogram (M)	III. $\sqrt{\frac{K^2L^2c^3}{Gh}}$
(D) Kelvin (K)	IV. $\sqrt{\frac{Gh}{c^3}}$

where h (Planck's constant), G (gravitational constant) and c (speed of light in vacuum) as fundamental units.

Choose the correct answer from the options given below:

(1) A-II, B-IV, C-I, D-III

(2) A-IV, B-II, C-I, D-III

(3) A-IV, B-I, C-II, D-III

(4) A-III, B-I, C-II, D-IV

Sol. $[h] = [ML^2T^{-1}]$

$[c] = [LT^{-1}]$

and $[G] = [M^{-1}L^3T^{-2}]$

then $\left[\sqrt{\frac{hc}{G}}\right] = \left[\frac{ML^2T^{-1}LT^{-1}}{M^{-1}L^3T^{-2}}\right]^{1/2} = [M]$

$$\left[\sqrt{\frac{Gh}{c^5}}\right] = \left[\frac{M^{-1}L^3T^{-2}ML^2T^{-1}}{L^5T^{-5}}\right]^{1/2} = [T]$$

$$\left[\sqrt{\frac{k^2L^2c^3}{Gh}}\right] = \left[\frac{k^2L^2L^3T^{-3}}{M^{-1}L^3T^{-2}ML^2T^{-1}}\right]^{1/2} = [k]$$

$$\left[\sqrt{\frac{Gh}{c^3}}\right] = \left[\frac{M^{-1}L^3T^{-2}ML^2T^{-1}}{L^3T^{-3}}\right]^{1/2} = [L]$$

$A \rightarrow IV, B \rightarrow II, C \rightarrow I, D \rightarrow III$

27. In an experiment to determine the resistance of a given wire using Ohm's law, the voltmeter and ammeter readings are noted as 10V and 5 A, respectively. The least counts of voltmeter and ammeter are 500 mV and 200 mA, respectively. The estimated error in the resistance measurement is _____Ω

(1) 0.25 (2) 2

(3) 2.5 (4) 0.18

Sol. $V = IR$

$$R = \frac{10}{5} = 2\Omega$$

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

$$\frac{\Delta R}{R} = \frac{500\times10^{-3}}{10} + \frac{200\times10^{-3}}{5}$$

$$\frac{\Delta R}{2} = 0.05 + 0.04$$

$\Delta R = 0.09 \times 2 = 0.18\ \Omega$

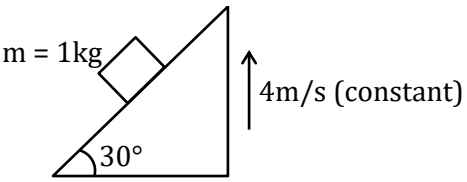
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28. A mass of 1 kg kept in a inclined plane with 30° inclination with respect to horizontal plane and it is at rest initially. Then the whole assembly is moved up with constant velocity of 4 m/s. The work done by the frictional force in time 2s is _____J.
(Take g = 10 m/s^2)

(1) 20 (2) 25

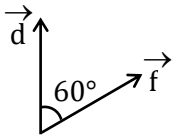
(3) 30 (4) 10

Sol.



f = mgsin30 = 5N

work done in 2 sec will be



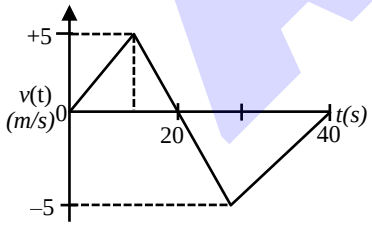
d = 4 × 2 = 8m

W = fd cos60°

W = (5)(8) cos60°

W = 20 joule

29. The velocity(v) versus time (t) plot of a particle is shown in the figure, for a time interval of 40s. The total distance travelled by the particle and the average velocity during this period are, respectively _____



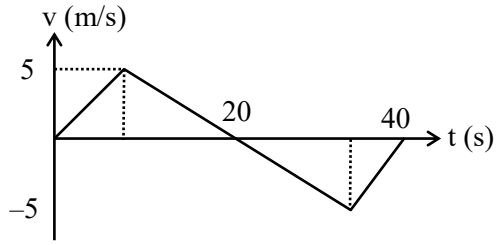
(1) 25 m and zero

(2) 50 m and zero

(3) 100 m and zero

(4) 100 m and 2.5 m/s

Sol.



Total distance travel

$d = \frac{1}{2}(20)(5)+\frac{1}{2}(20)\times 5$

d = 100 m

Average velocity = $\frac{\text{Total displacement}}{\text{Total time}}$

$\langle v\rangle = \frac{0}{40} = 0$

30. A wheel initially at rest is subjected to a uniform angular acceleration about its axis. In the first 2 s it rotates through an angle θ_1 and in the next 2 s it rotates through an angle θ_2. The ratio $\frac{\theta_2}{\theta_1}$ is_____.

(1) 6 (2) 3

(3) 4 (4) $\frac{1}{3}$

Sol. Let α be angular acceleration

In first 2 sec

$\theta_1 = \frac{1}{2}.\alpha(2)^2 = 2\alpha$

Angular covered in 4 sec

$\theta' = \frac{1}{2}(\alpha)(4)^2 = 8\alpha$

Angle in next 2 sec

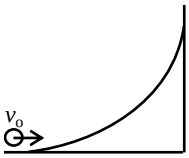
$\theta_2 = \theta' - \theta_1 = 8\alpha - 2\alpha$

$\theta_2 = 6\alpha$

$\frac{\theta_2}{\theta_1} = \frac{6\alpha}{2\alpha} = 3$

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31. An object of uniform density rolls up the curved path with the initial velocity v_0 as shown in the figure. If the maximum height attained by an object is $\frac{7v_0^2}{10g}$ (g = acceleration due to gravity), the object is a_____.

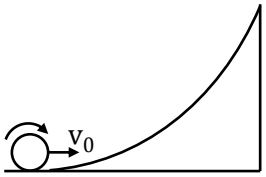


(1) solid cylinder (2) ring

(3) disc (4) solid sphere

Sol. $h = \frac{7v_0^2}{10g}$

Applying work energy theorem



$$-mgh = 0 - \frac{1}{2}mv^2 - \frac{1}{2}I\omega^2$$

$$-mg\left(\frac{7v_0^2}{10g}\right) = -\frac{1}{2}mv_0^2 - \frac{1}{2}\frac{Iv_0^2}{R^2}$$

$$\frac{7mv_0^2}{10} = \frac{mv_0^2}{2} + \frac{Iv_0^2}{2R^2}$$

$$\frac{mv_0^2}{5} = \frac{Iv_0^2}{2R^2}$$

$$I = \frac{2mR^2}{5}$$

32. A body of mass m is taken from the surface of earth to a height equal to twice the radius of earth (R_e). The increase in potential energy will be____.

(g is acceleration due to gravity at the surface of earth)

(1) $\frac{1}{2}mgR_e$ (2) $\frac{3}{4}mgR_e$

(3) $\frac{1}{4}mgR_e$ (4) $\frac{2}{3}mgR_e$

Sol. $U_1 = \frac{-GMm}{R_e}$

$$U_2 = \frac{-GMm}{R_e + 2R_e} = -\frac{GMm}{3R_e}$$

$$\Delta U = U_2 - U_1 = \frac{-GMm}{3R_e} + \frac{GMm}{R_e}$$

$$\Delta U = \frac{2}{3}\frac{GMm}{R_e} = \frac{2}{3}\left[\frac{GM}{R_e^2}\right](mR_e)$$

$$\Delta U = \frac{2}{3}mgR_e$$

33. Eight mercury drops, each of radius r, coalesce to form a bigger drop. The surface energy released in this process is_____. (S is the surface tension of mercury).

(1) $8\pi r^2S$ (2) $16\pi r^2S$

(3) $64\pi r^2S$ (4) $4\pi r^2S$

Sol. Volume conservation

$$(8)\left[\frac{4}{3}\pi r^3\right] = \frac{4}{3}\pi R^3$$

$R = 2r$

$U_i = (8)(4\pi r^2)(s)$; s = surface tension

$U_i = 32\pi r^2 s$

$U_f = (4\pi R^2)(s)$

$U_f = 4\pi(2r)^2(s) = 16\pi r^2 s$

Energy released = $U_i - U_f$

Energy released = $16\pi r^2 s$

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34. An ideal gas at pressure P and temperature T is expanding such that PT^3 = constant. The coefficient of volume expansion of the gas is____.

(1) $\frac{2}{T}$ (2) $\frac{1}{T}$

(3) $\frac{4}{T}$ (4) $\frac{3}{T}$

Sol. PT^3 = constant (C)

$$\frac{nRT}{V}.T^3 = C$$

$$V = \frac{nR}{C}.T^4$$

Volume expansion coeff.

$$\gamma = \frac{1}{V}\frac{dV}{dT}$$

$$\gamma = \frac{C}{nRT^4}.\frac{nR}{C}.4T^3 = \frac{4}{T}$$

35. Match List-I with List-II.

List-I		List-II	
A.	$\sin^2\omega t$	I.	Periodic with time period $T = \frac{\pi}{\omega}$ but not simple harmonic motion (SHM)
B.	$\sin^3(2\omega t)$	II.	Periodic with time period $T = \frac{2\pi}{\omega}$ but Not SHM
C.	$\sin(\omega t) + \cos(\pi\omega t)$	III.	periodic with time period $T = \frac{\pi}{\omega}$ and SHM
D.	$\cos\omega t + \cos 2\omega t$	IV.	Non-periodic

Choose the correct answer from the options given below.

(1) A-III, B-I, C-IV, D-II (2) A-II, B-I, C-III, D-IV

(3) A-III, B-II, C-IV, D-I (4) A-II, B-I, C-IV, D-III

Sol. (A) $\sin^2\omega t = \frac{1}{2} - \frac{1}{2}\cos 2\omega t$

it represents SHM motion having period = $\frac{\pi}{\omega}$

(B) $\sin^3 2\omega t = \frac{3}{4}\sin 2\omega t - \frac{1}{4}\sin 6\omega t$

common period $\frac{\pi}{\omega}$ but not SHM.

(C) $\sin\omega t + \cos\pi\omega t \rightarrow$ no common period hence it is a nonperiodic function

(D) $\cos\omega t + \cos 2\omega t$

common period $\frac{2\pi}{\omega}$ but it do not represent SHM.

A $\rightarrow$ III, B $\rightarrow$ I, C $\rightarrow$ IV, D $\rightarrow$ II

36. A metal rod of length L rotates about on end at origin with a uniform angular velocity ω. The magnetic field radially falls off as $B(r) = B_o e^{-\lambda r}$; λ being a positive constant. The emf induced (neglecting the centripetal force on electrons in the rod) is:

(1) $B_o\omega\left[\frac{1}{\lambda^2} - e^{-\lambda L}\left(\frac{1}{\lambda^2} + \frac{L}{\lambda}\right)\right]$

(2) $B_o\omega\left[\frac{1}{\lambda^2} + e^{-\lambda L}\left(\frac{1}{\lambda^2} + \frac{L}{\lambda}\right)\right]$

(3) $B_o\omega\left[\frac{4}{\lambda^2} - e^{-2\lambda L}\left(\frac{1}{\lambda^2} + \frac{2L}{\lambda}\right)\right]$

(4) $B_o\omega\left[\frac{3}{\lambda^2} - e^{-3\lambda L}\left(\frac{3}{\lambda^2} + \frac{L}{\lambda}\right)\right]$

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Sol. $e = \int_0^L B(\omega r)dr = B_0\omega\int_0^L e^{-\lambda r}.rdr$

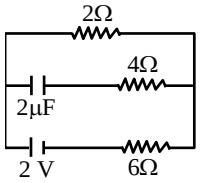
$= B_0\omega\left[r\int e^{-\lambda r}dr - \int 1.\frac{e^{-\lambda r}}{-\lambda}dr\right]_0^L$

$= B_0\omega\left[\frac{re^{-\lambda r}}{-\lambda} - \frac{e^{-\lambda r}}{\lambda^2}\right]_0^L$

$= B_0\omega\left[\left(0+\frac{1}{\lambda^2}\right) - \left(\frac{Le^{-\lambda L}}{\lambda} + \frac{e^{-\lambda L}}{\lambda^2}\right)\right]$

$e = B_0\omega\left[\frac{1}{\lambda^2} - e^{-\lambda L}\left(\frac{1}{\lambda^2}+\frac{L}{\lambda}\right)\right]$

37. Under steady state condition the potential difference across the capacitor in the circuit is____V.



(1) 0.5 (2) 1.5

(3) 0 (4) 2

Sol. $i = \frac{2}{6+2} = \frac{1}{4}A$

$v_C = v_{2\Omega} = i \times 2 = \frac{1}{2}v = 0.5\ V$

38. A particle of charge q and mass m is projected from origin with an initial velocity $\vec{v} = \left(\frac{v_0}{\sqrt{2}}\hat{x} + \frac{v_0}{\sqrt{2}}\hat{y}\right)$. There exists a uniform magnetic field $\vec{B} = B_0\hat{z}$ and a space varying electric field $\vec{E} = E_0e^{-\lambda x}\hat{x}$ within the region $0 \le x \le L$. After travelling a distance such that x-coordinate has changed from $x = 0$ to $x = L$, the change in the kinetic energy is_____.

(1) $\frac{qE_0}{\lambda}[1-e^{-\lambda L}]$

(2) $\left(\frac{v_0qB_0}{2\lambda}\right)[2-e^{-2\lambda L}]$

(3) $\frac{qE_0}{\lambda}[1+e^{-\lambda L}]$

(4) $q\left[\frac{E_0+v_0B_0}{\lambda}\right]\left[1-e^{-\lambda L/2}\right]$

Sol. $\Delta K = W_e + W_m = W_e + 0$

$= \int_0^L qEdx = qE_0\int_0^L e^{-\lambda x}dx$

$= \frac{qE_0}{\lambda}\left(1-e^{-\lambda L}\right)$

39. Given below are two statements : one is labelled as **Assertion (A) :** and the other is labelled as **Reason (R).**

Assertion (A) : The electromagnetic wave exerts pressure on the surface on which they are allowed to fall.

Reason (R) : There is no mass associated with the electromagnetic waves.

In the light of the above statements, choose the **correct answer** from the options given below :

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(2) Both **(A)** and **(R)** are true but **(R)** is **not** the correct explanation of **(A)**

(3) **(A)** is true but **(R)** is false

(4) **(A)** is false but **(R)** is true

Sol. Assertion and reason both are correct but not related.

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40. A thin convex lens and a thin concave lens are kept in contact and are co-axial. Which of the following statements is correct for this combination of two lenses?

(1) behaves as concave lens if $|f_{convex}| > |f_{concave}|$

(2) behaves as concave lens if $|f_{convex}| < |f_{concave}|$

(3) behaves as convex lens if $|f_{convex}| > |f_{concave}|$

(4) Focal length of the lens system will change if the position of two lenses are interchanged

Sol. $1 \rightarrow$ convex lens

$2 \rightarrow$ concave lens

$$\frac{1}{f}=\frac{1}{f_1}+\frac{1}{-f_2}:\frac{1}{f_1}-\frac{1}{f_2}$$

for $f_1 > f_2 \therefore f < 0$

$\therefore$ concave lens.

41. An object AB is placed 15 cm on the left of a convex lens P of focal length 10 cm. Another convex lens Q is now placed 15 cm right of lens P. If the focal length of lens Q is 15 cm, the final image is_____

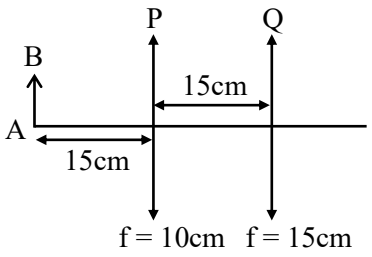
(1) virtual, formed at 7.5 cm right of lens Q, with a size bigger than that of AB

(2) real, formed at 7.5 cm right of lens Q, with a size same than that of AB

(3) formed at infinity.

(4) real, formed at 7 cm right of lens Q, with a size smaller than that of AB

Sol.



for P $\frac{1}{V}-\frac{1}{-15}=\frac{1}{10}$

$\therefore$ v = +30 cm = +15 cm right of Q.

for Q $\frac{1}{V'}-\frac{1}{15}=\frac{1}{15}$

V' = 7.5 cm

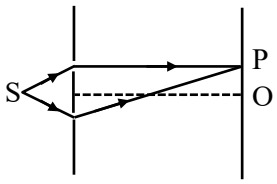
$$m = m_1m_2 = \frac{30}{-15}\times\frac{7.5}{15} = -1$$

42. The maximum intensity in a Young's double slit experiment is I_0. Distance between the slits (d) is 5λ, where λ is the wavelength of light used. The intensity of the fringe, exactly opposite to one of the slits on the screen, placed at D = 10 d is____

(1) $\frac{I_0}{4}$ (2) $\frac{I_0}{2}$

(3) I_0 (4) $\frac{3I_0}{4}$

Sol.



At P $y = \frac{d}{2}$

$$\Delta Q = \frac{2\pi}{\lambda}\frac{yd}{D} = \frac{2\pi}{\lambda}\frac{d}{2}\frac{d}{D} = \frac{\pi d^2}{\lambda D}$$

$$\Delta Q = \frac{\pi\times 25\lambda^2}{\lambda\times 10d} = \frac{5\pi\lambda}{2\times 5\lambda} = \frac{\pi}{2}$$

$$I = I_0\cos^2\left(\frac{\Delta Q}{2}\right) = I_0\cos^2\left(\frac{\pi}{2\times 2}\right) = \frac{I_0}{2}$$

43. An electron is travelling with a velocity *v* in free space and when it enters a medium, its velocity is reduced by 20%. The de Broglie wavelength of electron in the medium is $\alpha\lambda_0$, where λ_0 is its de Broglie wavelength in free space. The value of α is____

(1) 1.20 (2) 1.0

(3) 1.25 (4) 0.75

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Sol. $\lambda_0 = \frac{h}{mv_0}$

$\lambda = \frac{h}{mv} = \frac{h}{m(0.8v_0)} = \frac{1}{0.8}\frac{h}{mv_0} = 1.25\,\lambda_0$

$\therefore\ \alpha = 1.25$

44. Assuming the experimental mass of $^{12}_{6}C$ as 12u, the mass defect of $^{12}_{6}C$ atom is ____ MeV/c^2. (Mass of proton = 1.00727u. mass of neutron = 1.00866 u, 1 u = 931.5 MeV/c^2 and c is the speed of the light in vacuum.)

(1) 127.5 (2) 89.03

(3) 272.0 (4) 92.0

Sol. $\Delta m = (6m_p + 6m_n) - m_C$

$= 6(1.00727 + 1.00866) - 12$

$= 0.09558$ amu $= 89.03$ MeV/C^2

45. In a semiconductor p-n diode, the doping concentrations on p-side and n-side are 10^{15} atoms/cm^3 and 10^{18} atoms/cm^3 , respectively. Which one of the following statement is true?

(1) Widths of depletion region on either side of the interface are equal

(2) The depletion region width is more on p-side compared to that in n-side

(3) The depletion region width is more on n-side compared to that in p-side

(4) No depletion region forms because of unequal doping concentrations on p and n-sides

Sol. $N_p x_p = N_n x_n$

$N_p < N_n\ \therefore\ x_p > x_n$ in depletion region

SECTION-B

46. A copper wire of length 3 m is stretched by 3 mm by applying an external force. The volume of the wire is 600×10^{-6} m^3. The elastic potential energy stored in the wire in stretched condition would be _____J.

(Given Young modulus of copper = 1.1×10^{11} N/m^2)

Sol. $U = \frac{1}{2}y(\text{strain})^2 \times \text{volume}$

$= \frac{1}{2} \times 1.1 \times 10^{11} \times \left(\frac{3 \times 10^{-3}}{3}\right)^2 \times 600 \times 10^{-6}$

$U = 33$ J

47. The heat extracted out of x gram of water initially at 50°C to cool it down to 0°C is sufficient to evaporate (1000 – x) gram of water also initially at 50°C. The value of x (closest integer) is____.

(Take latent heat of water 2256 kJ/kg. K, specific heat capacity of water 4200 J/kg. K)

Sol. $x \times 4200 \times 50 = (1000 - x) \times 4200 \times 50 + (1000 - x) \times 2256 \times 10^3$

$x = 1000 - x + (1000 - x) \times 10.74$

$x = 921.52$

48. A series LCR circuit with R = 20Ω, L = 1.6 H and C = 40 μF is connected to a variable frequency a.c. source. The inductive reactance at resonant frequency is _____Ω.

Sol. For resonance

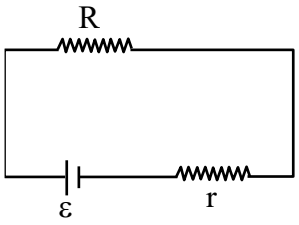
$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1.6 \times 40 \times 10^{-6}}} = \frac{10^3}{8} = 125$

$X_L = \omega L = 125 \times 1.6 = 200\ \Omega$

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49. When an external resistance of 5Ω is connected across terminals of a cell, a current of 0.25 A flows through it. When the 5Ω resistor is replaced by a 2Ω resistor, a current of 0.5 A flows through it. The internal resistance of the cell is ____Ω.

Sol.



$$i = \frac{\varepsilon}{R+r}$$

$$0.25 = \frac{\varepsilon}{5+r}$$

$$0.5 = \frac{\varepsilon}{2+r}$$

on solving, $r = 1\Omega$

50. A circular loop of radius 20 cm and resistance 2Ω is placed in a time varying magnetic field $\vec{B} = \left(2t^2 + 2t + 3\right)T$. At t = 0, for the plane of the loop being perpendicular to the magnetic field and, the induced current in the loop at t = 3 s is $\frac{\alpha}{50}A$. The value of α is_____ (Take π = 22/7)

Sol. $\phi = B \cdot \pi r^2$

$$\varepsilon = \frac{d\phi}{dt} = \pi r^2 \frac{dB}{dt}$$

$$i = \frac{\varepsilon}{R} = \frac{\pi r^2}{R}.\left(\frac{dB}{dt}\right) = \frac{\pi}{2}\left(\frac{1}{5}\right)^2 (4t+2)$$

for t = 3

$$i = \frac{11}{7} \times \frac{1}{25} \times 14$$

$$i = \frac{44}{50}$$

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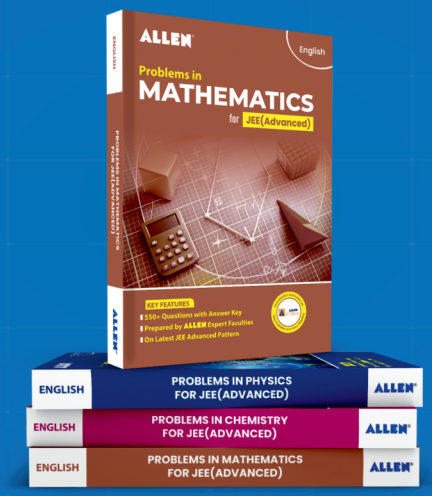


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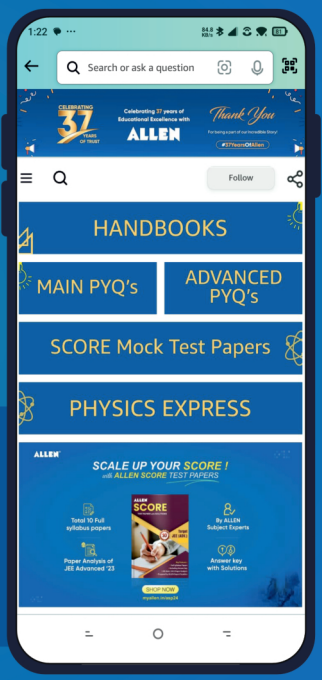
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